CLEMENT CHUNG

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Education

University of Waterloo **Waterloo, ON**
Bachelor of Computer Science, Sep 2025 – May 2030
- cGPA: 4.0 + B.P. Dammizio Entrance Scholarship

Skills

Languages: C++, Rust, CUDA, Python, Java, TypeScript
Systems/Tools: Linux, CMake, cuBLAS, Docker, Git, WebRTC

Experience

Systems Software Intern **May 2026 – Present**
Martinrea International *Vaughan, ON*
- Implemented **Python environment caching**, reusing virtual environments across projects to cut CI test time by **20%** and build time by **66%** for a tool used by **85** engineers.
- Fixed **flaky Linux CI tests and data races**, including bugs in process and signal handling, bringing the affected integration test to a **100% observed pass rate**
- Built a C++ device whitelist with subnet filtering, live updates, and IPv4/IPv6 support across REST and WebSocket connections

Real-Time Voice AI Intern **Jul 2025 – Aug 2025**
AI Stealth Startup *Mountain View, CA*
- Architected a latency-sensitive, dual-agent streaming pipeline that isolated retrieval and API workloads from response generation, reducing end-to-end response latency by **6× (1500 ms → 250 ms)**.
- Built a real-time lip-sync engine that processed **WebRTC audio at 10 ms intervals**, replacing ML-based phoneme detection to achieve near-zero animation latency.

Projects

MiniGPT (LLM from Scratch) | Rust, C++, CUDA
- Built a GPT-style LLM from scratch with a Rust frontend and C++/CUDA backend for GPU-accelerated inference and training without machine learning frameworks
- Optimized CUDA kernels with coalesced memory access, shared-memory tiling, bank-conflict avoidance, and tensor core utilization, for up to **10x faster performance** than naive implementations
- Optimized a custom **32K-token BPE tokenizer in Rust** by updating only affected token pairs during each merge instead of rescanning the full sequence, encoding 32K tokens in **46** seconds

DingleBob (Small Interpreter) | Rust
- Built a **small interpreter** with lexical scoping, closures, multi-file imports, and REPL/file execution
- Added built-in system functions for file I/O, environment variables, and timing, plus error messages that point directly to the failing source code

Awards

- **Hack the 6ix 2026 — QNX Best Use of QNX + Top 8 Finalist:** Built an autonomous turret with camera tracking and UART motor control on a QNX rasberry pi
- **Hack Canada 2026 — Most Technically Complex AI Hack:** Built an AI tool that turns math questions into interactive visualizations

Activities

Waterloo Virtual Reality Labs - Firmware Team Member | Oct 2025 - Present
- Worked on Linux display drivers and Monado/OpenXR integration on a Raspberry Pi 5 for a VR OS.

FRC Team 2198 — Programming Lead | Sep 2021 – Apr 2025
- Built robot software for AprilTag localization, autonomous alignment, pose estimation, and sensor integration across multiple competition seasons